Global null control of a two-phase Stefan interface with signed initial temperatures

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October 12, 2026

Abstract

We prove that two Dirichlet boundary controls can drive a one-dimensional two-phase Stefan system to zero temperature and any prescribed interior interface position in every positive time. The initial temperatures may be arbitrary real smooth functions compactly supported in their phases, and the interface remains strictly inside the rod throughout. A short uncontrolled evolution gives analytic interface traces without a sign assumption. These traces are then extended by Gevrey interpolation and the flatness series of Colle, Lohéac and Takahashi, while a convex interpolation of the interface position prevents collision with the endpoints.

1 The control problem

In the two-phase Stefan system, heat diffusion acts on both sides of a moving interface, and the difference of the interface heat fluxes determines its velocity. Boundary-temperature control asks whether the two exterior temperatures can be chosen to bring both phases to zero temperature and place the interface at a prescribed point. The geometric constraint is essential: neither phase may disappear during the control.

Colle, Lohéac and Takahashi [1, Theorem 1.1] prove arbitrary-time boundary controllability by a flatness construction for initially phase-signed temperatures, with strict signs at the exterior endpoints. The coefficient estimates in their Proposition 3.4 and Corollary 3.6 do not require these signs or a smallness condition. The signs enter their preparatory regularization argument. We instead prove short-time analyticity directly for arbitrary signed compactly supported profiles. This permits the same series reconstruction to start from the actual positive-time solution.

The target here is the unconstrained-sign formulation in AIM Problem 258 [4]. The temperature fields are treated as real solutions of the specified heat and Stefan equations; no phase-sign inequality is added. Let L denote the rod length, d_l, d_r the diffusivities and λ the latent-heat coefficient. All are positive. Let L_0, L_T be the initial and desired interface positions. Time is the first argument of every temperature below.

Theorem 1. *For every $L, d_l, d_r, \lambda, T > 0$, every $L_0, L_T \in (0, L)$, and every real $y_0 \in C_c^\infty(0, L_0)$ and $z_0 \in C_c^\infty(L_0, L)$, there exist $v_l, v_r \in L^\infty(0, T)$, $\ell \in C^1([0, T])$ with $0 < \ell < L$, and temperatures satisfying*

$$y_t = d_l y_{xx} \quad (0 < x < \ell(t)), \qquad z_t = d_r z_{xx} \quad (\ell(t) < x < L),$$

$$y(t, 0) = v_l(t), \quad z(t, L) = v_r(t), \quad y(t, \ell(t)) = z(t, \ell(t)) = 0,$$

$$d_l y_x(t, \ell(t)^-) - d_r z_x(t, \ell(t)^+) = -\lambda \ell'(t).$$

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The initial data and interface are (y_0, z_0, L_0), the terminal temperatures vanish, and $\ell(T) = L_T$. Zero extensions of both temperatures must belong to $C([0,T]; L^2(0,L))$. No temperature sign, smallness condition, or uniform control-cost bound is imposed.

The construction gives a classical solution for positive time, with continuous temperatures and spatial first derivatives in phase coordinates down to zero. All constants may depend on the entire parameter and initial-data tuple.

The result allows arbitrarily large data and arbitrarily small control times; it supplies no common bound on the control amplitudes. The controlled temperatures need not retain phase signs. The initial class remains smooth and compactly supported, and no higher-dimensional statement is made.

The proof first solves the system for a short time on two fixed intervals. Complex-time heat estimates give an analytic interface and analytic flux traces at positive time. The PDE then supplies spatial analyticity, so the actual solution agrees with its interface-centered flatness series. We keep this agreement on an open time interval and cut off the traces afterwards. The interface is a convex combination of its short-time path and the target position, which keeps both phases nonempty.

The following sections give the normalization, the short-time construction, the series reconstruction, and the final control. Complex scalars occur only in the analyticity argument; the final solution and controls are real.

2 Normalization

Put $r = x/L$, $b = \ell/L$, $b_0 = L_0/L$, $b_T = L_T/L$, and

$$c_- = d_l/L^2, \qquad c_+ = d_r/L^2, \qquad \theta_-(t,r) = -\frac{d_l}{\lambda L^2} y(t, Lr), \qquad \theta_+(t,r) = -\frac{d_r}{\lambda L^2} z(t, Lr).$$

Then $\theta_{\pm,t} = c_\pm \theta_{\pm,rr}$ in their phases, the interface values vanish, and

$$b' = \theta_{-,r}(t, b^-) - \theta_{+,r}(t, b^+). \tag{1}$$

Indeed the right side is $[-d_l y_x + d_r z_x]/(\lambda L) = \ell'/L$. Conversely these identities recover exactly the physical equations and the controls by multiplying each phase by $-\lambda L^2/d_\pm$. It therefore suffices to work on $[0,1]$ with arbitrary $c_\pm > 0$.

3 Short-time solutions and analyticity

Lemma 2 (Spectral estimates)**.** *Let $e_n(s) = \sqrt{2}\sin(n\pi s)$ and $\mu_n = (n\pi)^2$. For $0 \le \gamma$, set*

$$E_\gamma = \{u = (u_n) : \ \|u\|_{E_\gamma}^2 = \sum_{n\ge 1} \mu_n^{2\gamma} |u_n|^2 < \infty\}.$$

These are complex Banach spaces. Identify E_0 with the completion of the sine polynomials under the integral L^2 norm. Every continuous function on $[0,1]$ is represented in this completion by its sine coefficients. For $\alpha = 7/8$, E_α embeds continuously into $C^1([0,1])$, with zero endpoint values. For $\Re z > 0$, define

$$S(z)u = (e^{-\mu_n z} u_n)_{n\ge 1}, \qquad Au = (\mu_n u_n)_{n\ge 1}, \quad D(A) = E_1.$$

If $\Re z \ge k|z|$ for a fixed $k > 0$, then

$$\|S(z)\|_{E_\alpha \to E_\alpha} \le 1, \quad \|S(z)\|_{E_0 \to E_\alpha} \le C_k |z|^{-\alpha}, \quad \left\|S'(z)\right\|_{E_0 \to E_\alpha} \le C_k |z|^{-\alpha-1}. \tag{2}$$

Moreover $S(z)$ is holomorphic away from zero, strongly continuous at zero within such a sector, and if $u \in E_{\alpha+1}$, $\|(S(z) - S(w))u\|_{E_\alpha} \le |z-w| \, \|Au\|_{E_\alpha}$ whenever the segment from z to w lies in $\Re z \ge 0$.

Proof. Completeness of each weighted sequence space follows by taking coordinate limits of a Cauchy sequence: finite partial sums pass to the limit, and the Cauchy tail estimate bounds the full norm of each difference. Direct integration gives $\int_0^1 e_m e_n \,\mathrm{d}s = \delta_{mn}$. For a continuous function, odd reflection about zero followed by 2-periodic extension is bounded and piecewise continuous, hence Riemann integrable. Apply [3, Theorem 11.8.13] after the change of variable $s \mapsto \pi s$: the cosine coefficients vanish and the sine partial sums converge in $L^2(0,1)$, with squared norm equal to the sum of squared coefficients. This proves the asserted representation and its isometry on continuous functions. No endpoint condition is required for this L^2 assertion.

For $u \in E_\alpha$, Cauchy–Schwarz for finite sums, followed by the convergent numerical series $\sum n^{2-4\alpha}$, gives

$$\sum n|u_n| \le C \|u\|_{E_\alpha}.$$

Both the sine series and its derivative thus converge uniformly, proving the embedding and endpoint statement. Inequalities (2) follow from $\sup_{v\ge 0} v^p e^{-kv} < \infty$ for $p = \alpha, \alpha+1$. Finite spectral sums are holomorphic; their operator derivatives converge locally uniformly away from zero by the same bounds. Strong continuity at zero follows by splitting the sequence into a finite initial segment and a uniformly small tail. Finally integrate $S'(\zeta)u = -S(\zeta)Au$ along the indicated segment. Compactly supported smooth functions belong to every E_γ, by repeated integration by parts in their sine coefficients.

The scalar Cauchy formula and its Taylor, estimate and uniform-limit consequences [2, Theorems 3.2.15, 3.3.1, 3.3.8 and 3.4.5] also hold for functions valued in these sequence spaces or their finite products. The vector Cauchy formula follows by Riemann integration on a circle and the scalar formula in each coordinate. The integral's norm is bounded by length times the supremum norm. Coordinate equality identifies the vectors. This proves the vector Cauchy estimate and power expansion. The same formula proves that a locally uniform vector limit of holomorphic functions is holomorphic. An improper vector integral dominated in norm by an integrable scalar function exists by the Cauchy criterion; if its truncated integrals are holomorphic, the preceding limit argument applies. □

Lemma 3 (Local existence and time analyticity)**.** *For every $c_\pm > 0$, $b_0 \in (0,1)$ and real smooth compactly supported initial profiles in the two phases, there is $\delta > 0$ and a solution with zero exterior controls on $[0,\delta]$. Its interface is C^1, stays in $(0,1)$, and in fixed phase coordinates the profiles are continuous into E_α down to zero and holomorphic in a complex time sector at every positive real time. The interface and both interface slopes are analytic in time on $(0,\delta)$.*

Proof. Write

$$Y(t,s) = \theta_-(t, b(t)s), \qquad Z(t,s) = \theta_+(t, b(t) + (1-b(t))s).$$

For $U = (Y, Z, b)$ define

$$F(U) = Y_s(1)/b - Z_s(0)/(1-b), \quad a_-(b) = c_-/b^2, \quad a_+(b) = c_+/(1-b)^2,$$

$$G_-(U) = \frac{F(U)}{b} s Y_s, \qquad G_+(U) = \frac{F(U)}{1-b}(1-s)Z_s.$$

The transformed system is

$$b' = F(U), \quad Y_t = -a_-(b)AY + G_-(U), \quad Z_t = -a_+(b)AZ + G_+(U). \tag{3}$$

The two signs of the drift follow by differentiating the two coordinate maps at fixed s. Let $U_0 = (Y_0, Z_0, b_0)$. Lemma 2 makes F and $G_\pm$ holomorphic and locally Lipschitz, respectively into $\mathbb{C}$ and E_0, on $E_\alpha^2 \times \mathbb{C}$ where $b \ne 0, 1$. Continuous spatial products use the isometry established in Lemma 2.

Choose $\vartheta = \pi/6$. On the closed time sector $K_\delta = \{t : |t| \le \delta,\ |\arg t| \le \vartheta\}$, including its tip, use the supremum product norm. Consider continuous functions, holomorphic in the

sector interior, with $U(0) = U_0$, $\|Y - Y_0\|_{E_\alpha}, \|Z - Z_0\|_{E_\alpha} \le 1$ and $|b - b_0| \le r$. Choose a positive r sufficiently small that $b, 1-b$ are nonzero and $|a_\pm(b) - a_\pm(b_0)| \le \varepsilon a_\pm(b_0)$ throughout this complex disk, with a fixed sufficiently small $\varepsilon > 0$. This is a closed subset of a complete supremum space: completeness and closure of holomorphy follow from Lemma 2. The rational functions and their derivatives are uniformly bounded on this set.

Set $\tau_\pm(t) = \int_0^t a_\pm(b(q))\,\mathrm{d}q$, using straight segments. For $q = \rho t$, $0 \le \rho < 1$, one has

$$\tau_\pm(t) - \tau_\pm(q) = t \int_\rho^1 a_\pm(b(ut))\,\mathrm{d}u.$$

Consequently these differences lie in a fixed closed sector of the right half-plane and their modulus is comparable to $(1-\rho)|t|$. The same is true of the straight segment joining these increments for two inputs. Their difference is at most $C(1-\rho)|t| \left\|b - \widehat{b}\right\|_\infty$.

Define

$$\Phi_b(U)(t) = b_0 + \int_0^t F(U(q))\,\mathrm{d}q,$$

$$\Phi_Y(U)(t) = S(\tau_-(t))Y_0 + \int_0^t S(\tau_-(t) - \tau_-(q))G_-(U(q))\,\mathrm{d}q, \tag{4}$$

and the analogous expression for Φ_Z. Lemma 2 gives a bound $C\delta^{1-\alpha}$ for each integral in (4). It is continuous at the tip with value zero. Holomorphy follows by writing it as $t\int_0^1 \cdots \mathrm{d}\rho$: truncations away from $\rho = 1$ are holomorphic (Riemann sums and the vector Cauchy formula), and the tail is locally uniformly bounded by $C\int(1-\rho)^{-\alpha}\,\mathrm{d}\rho$. The primitives starting at the tip can be justified by starting at $\rho = \epsilon$ and letting $\epsilon \downarrow 0$, with bounded tails.

For two inputs, integration of the S' estimate in (2) on the straight segment joining their increments proves

$$\|S(\tau(t) - \tau(\rho t)) - S(\widehat{\tau}(t) - \widehat{\tau}(\rho t))\|_{E_0 \to E_\alpha} \le C \left\|b - \widehat{b}\right\|_\infty [(1-\rho)|t|]^{-\alpha}.$$

The initial terms differ by at most $C\delta \left\|b - \widehat{b}\right\|_\infty \|AY_0\|_{E_\alpha}$, and similarly for Z_0. Thus

$$\left\|\Phi(U) - \Phi(\widehat{U})\right\|_\infty \le C(\delta + \delta^{1-\alpha}) \left\|U - \widehat{U}\right\|_\infty. \tag{5}$$

The same estimates against U_0 show that Φ maps the indicated closed set into itself for sufficiently small δ: the scalar change is $O(\delta)$, and each profile change is $O(\delta \|AY_0\|_{E_\alpha} + \delta^{1-\alpha})$. All constants are finite for these fixed, possibly large, data. Choose δ also making (5) less than one. Iteration converges, since successive differences are bounded by a summable geometric series in a complete space. Passing to the limit in (4) gives a fixed point; (5) gives uniqueness within this set. Conjugation preserves the map and the initial data, so the fixed point is real on the real axis.

At positive real time, its E_α derivatives exist by holomorphy. Each sine coefficient in (4) solves $Y_n' = -\mu_n a_- Y_n + (G_-)_n$. It follows that $(\mu_n Y_n)$ belongs to E_0, because $a_- AY = G_- - Y_t \in E_0$; similarly for Z. The identity $AY = -Y_{ss}$ holds in distributions by testing against smooth compactly supported functions and passing in the sine partial sums. As Y_t and G_- are continuous in space, this second derivative is continuous. To see the classical conclusion directly, integrate the continuous right side twice; the difference from Y has zero distributional second derivative and hence is affine. One elementary justification of this last assertion is to convolve on every smaller open interval with a smooth, compactly supported mollifier of integral one. The convolutions have zero ordinary second derivative and are affine there; they converge uniformly on compact intervals to the continuous difference. A uniform limit of affine functions is affine, as its values at two distinct points determine the coefficients. Differentiation of this convolution

is legitimate because the mollifier and every derivative have compact support and bounded integrands. Therefore (3) is classical. The same reasoning applies to all time derivatives. Endpoint values are zero by Lemma 2. The scalar equation gives $b \in C^1$ down to zero, with $b'(0) = 0$ since both initial profiles vanish near the interface. The E_α traces and the rational factors show that $q_- = Y_s(1)/b$ and $q_+ = Z_s(0)/(1-b)$ are analytic in time. Inverting the coordinate maps proves all claimed local properties. □

Lemma 4 (Spatial analyticity)**.** *On every compact $J \subset (0,\delta)$, the local profiles of Lemma 3 are analytic in space up to both phase endpoints. Their Taylor series at the interface represent the actual profiles in an interface neighborhood.*

Proof. Cauchy estimates in E_α and Lemma 2 give, for $q = 0, 1$, $\sup_{J\times[0,1]} |\partial_t^n \partial_s^q Y| \le C A_0^n n!$. Equation (3) rearranges as

$$Y_{ss} = a(t) Y_t + h(t) s Y_s, \quad a = b^2/c_-, \quad h = -bb'/c_-. \tag{6}$$

For Z, replace a by $(1-b)^2/c_+$ and hs by $-(1-b)b'(1-s)/c_+$. On a slightly larger compact interval these coefficients have bounds $M_a A_0^n n!$, $M_h A_0^n n!$ by Lemma 3 and [2, Theorem 3.3.8]. Equation (6) and its time derivatives bootstrap continuous spatial derivatives of every order: at each step its right side is continuous, and the preceding twice-integration argument applies. Time differentiation is justified first in distributions, where Lemma 3 provides all E_α time derivatives; the resulting continuous representatives coincide with the classical derivatives.

Choose $A \ge 2A_0$. Induction on q, simultaneously for all n, gives

$$\sup_{J\times[0,1]} |\partial_t^n \partial_s^q Y| \le C A^n B^q (n+q)!. \tag{7}$$

For $q \ge 2$, use $\partial_s^{q-2}(sY_s) = s\partial_s^{q-1} Y + (q-2)\partial_s^{q-2} Y$. In each Leibniz sum, for $m \ge n$,

$$\sum_{k=0}^{n} \binom{n}{k} k!(A_0/A)^k (m-k)! \le 2m!,$$

because $\binom{n}{k} k!(m-k)!/m! \le 1$ and $A_0/A \le 1/2$. The ratios of the three resulting bounds to the right side of (7) are bounded respectively by

$$\frac{2M_a A}{B^2(n+q)}, \qquad \frac{2M_h}{B(n+q)}, \qquad \frac{2M_h(q-2)}{B^2(n+q)(n+q-1)}.$$

Their sum is at most one after increasing B, independently of n, q. The right-phase calculation has a minus sign in its last term and the same absolute bound. Setting $n = 0$ in (7), Taylor's remainder is bounded by $C(B|s-s_0|)^{q+1}$, proving the assertion also at one-sided endpoints. The remainder formula follows by repeated integration of the $(q+1)$st derivative along the spatial segment. □

4 Reconstruction from the interface traces

For $M, R > 0$ and $\sigma \ge 1$, write $G(M, R, \sigma)$ for the real $C^\infty(\mathbb{R})$ functions satisfying

$$\sup_t |g^{(n)}(t)| \le M (n!)^\sigma R^{-n} \qquad (n \ge 0).$$

A function is bounded Gevrey-σ if it belongs to one such class. For $\sigma = 1$ these derivative bounds imply analyticity; for $\sigma > 1$ they allow smooth cutoffs that are flat at their endpoints.

Lemma 5 (Flatness reconstruction)**.** *Fix $c > 0$, $1 \le \sigma < 2$ and globally bounded Gevrey-σ functions f, p on $\mathbb{R}$. Set $\beta_0 = 0$, $\alpha_0 = p$, and recursively*

$$\beta_{i+1} = c^{-1}(\beta_i' - f\alpha_i), \qquad \alpha_{i+1} = c^{-1}(\alpha_i' - f\beta_{i+1}). \tag{8}$$

Then

$$S(t,X) = \sum_{i\ge 0} \alpha_i(t) \frac{X^{2i+1}}{(2i+1)!} + \sum_{i\ge 0} \beta_i(t) \frac{X^{2i}}{(2i)!} \tag{9}$$

converges on every bounded spatial set with all time and space derivatives. It is entire in X, smooth in (t,X), and satisfies $S_t = cS_{XX} + fS_X$, $S(t,0) = 0$, $S_X(t,0) = p(t)$. If f, p and all their derivatives vanish at a time, then S vanishes identically in space at that time.

Proof. Choose a common $R > 0$ and positive upper constants such that $f \in G(M_f, R, \sigma)$, $p \in G(M_\alpha, R, \sigma)$ and $\beta_0 = 0 \in G(M_\beta, R, \sigma)$. Put $H = M_\beta/R + M_\alpha M_f$. With the source parameter $\chi = c^{-1}$, [1, Proposition 3.4] permits every $\rho > 0$ satisfying

$$\rho \le \frac{2^{3-\sigma}R}{3}, \qquad \rho \le \frac{2M_\alpha c}{3HM_f}.$$

Both upper bounds are strictly positive for any fixed input bounds. For all integers $k, i \ge 0$, the proposition then gives

$$\left\|\alpha_i^{(k)}\right\|_\infty \le \frac{M_\alpha c^{-i}}{R^k \rho^i} \frac{((k+2i)!)^\sigma}{(i!)^{2-\sigma}((2i)!)^{\sigma-1}},$$
$$\left\|\beta_{i+1}^{(k)}\right\|_\infty \le \frac{Hc^{-(i+1)}}{R^k \rho^i} \frac{((k+2i+1)!)^\sigma}{(i!)^{2-\sigma}((2i+1)!)^{\sigma-1}}.$$

For fixed time derivative order, the coefficient estimates of Proposition 3.4 leave a factor $(i!)^{-(2-\sigma)}$ after division by the spatial factorial, up to a fixed power of i and a geometric factor. This proves convergence even for complex X on any disk, and the same bound applies to every fixed mixed derivative. [1, Corollary 3.6] permits termwise differentiation. Coefficients of X^{2i} and X^{2i+1} in $cS_{XX} + fS_X$ are $c\beta_{i+1} + f\alpha_i = \beta_i'$ and $c\alpha_{i+1} + f\beta_{i+1} = \alpha_i'$. This proves the PDE. The interface identities follow from $\beta_0 = 0$ and $\alpha_0 = p$. Each coefficient uses finitely many jets of f, p by (8), proving the final assertion. The cited coefficient estimates impose no input-sign condition. □

Lemma 6 (Identification)**.** *Suppose on an open time interval in $(0, \delta)$ that $f = b'$ and $p = q_\pm$, and use (9) with $c = c_\pm$. Then (9), translated by $X = r - b(t)$, equals the corresponding local temperature throughout its closed spatial phase.*

Proof. For the local temperature expressed in X, the equation is $S_t = cS_{XX} + b'S_X$. Its even and odd spatial derivatives at zero must therefore obey (8), starting with value zero and slope $q_\pm$. All mixed derivatives exist by Lemma 3–Lemma 4, so this coefficient calculation is legitimate. Lemma 4 identifies the local profile with its Taylor series near zero, while Lemma 5 gives the same Taylor coefficients. Both profiles are real analytic throughout the connected phase interval, hence coincide everywhere. For completeness, the latter identity principle needs no extra PDE theorem: the set of points with a zero neighborhood for their difference is nonempty and open; at a limit point all derivatives vanish by continuity, so its convergent Taylor expansion gives a zero neighborhood. It is thus also closed in a connected interval. Endpoints follow by continuity. □

5 The global control

Products of bounded Gevrey-σ functions, affine time rescalings, and their derivatives are again bounded Gevrey-σ. Indeed Leibniz gives $\binom{n}{k}(k!)^\sigma((n-k)!)^\sigma \le (n!)^\sigma$; the $(n+1)$ terms and geometric factors are absorbed by decreasing R. Differentiation uses $(n+1)^\sigma \le C^n$ after adjusting the constant for $n = 0$. A real analytic function on a neighborhood of a compact interval has Cauchy bounds $MA^n n!$ there. Multiplying it by a Gevrey cutoff flat at the interval endpoints, then extending by zero, gives a globally bounded Gevrey-σ function by the same product bound. These closure properties are used below.

Lemma 7 (Normalized control construction)**.** *For all normalized data above and every $T > 0$, there are bounded Dirichlet controls driving both temperatures to zero and b to $b_T \in (0,1)$, with a C^1 interface strictly inside the rod.*

Proof. Let $(\widetilde{\theta}_\pm, \widetilde{b})$ be the solution of Lemma 3, with analytic slopes $\widetilde{q}_\pm$. Set $a = \min(\delta, T)/4$ and $t_* = a/2$. Fix $1 < \sigma < 2$. By [1, Lemma 3.2], choose a globally bounded Gevrey-σ function $\phi_\sigma : \mathbb{R} \to [0,1]$, equal to 1 for $t \le 0$ and 0 for $t \ge 1$. All its positive-order derivatives vanish at both 0 and 1. Define

$$\chi(t) = \phi_\sigma((t-a)/a), \qquad \eta(t) = 1 - \phi_\sigma((t-a/4)/(a/4)).$$

Thus $\chi = 1$ for $t \le a$, $\chi = 0$ for $t \ge 2a$, and $\eta = 0$ for $t \le a/4$, $\eta = 1$ for $t \ge t_*$. Define on $[a/4, 2a]$, and extend by zero outside,

$$f = \eta[\chi \widetilde{b}' + \chi'(\widetilde{b} - b_T)], \qquad p_- = \eta\chi\widetilde{q}_-, \qquad p_+ = p_- - f. \tag{10}$$

The local quantities are analytic on a neighborhood of this compact interval because $2a < \delta$. Flatness of the cutoff jets proves that these zero extensions are globally bounded Gevrey-σ functions. In particular there is no regularity claim at the original time zero for the uncontrolled solution.

For $t \ge t_*$, prescribe

$$b(t) = \begin{cases} \chi(t)\widetilde{b}(t) + (1-\chi(t))b_T, & t_* \le t \le 2a, \\ b_T, & t \ge 2a. \end{cases} \tag{11}$$

Use $\widetilde{b}$ for $0 \le t \le t_*$. On $[t_*, a]$ these definitions agree, together with their derivatives. Both entries in the convex combination (11) lie strictly in $(0,1)$, and $0 \le \chi \le 1$; hence so does b at every time. The flat jets at $2a$ give a smooth joining to the constant path. For $t \ge t_*$ one has $b' = f$. On the overlap $[t_*, a]$, $p_\pm = \widetilde{q}_\pm$, by (1).

Apply Lemma 5 separately with c_-, p_- and c_+, p_+, using the same f, and define for $t \ge t_*$

$$\theta_-(t,r) = S_-(t, r - b(t)), \quad 0 \le r \le b(t), \qquad \theta_+(t,r) = S_+(t, r - b(t)), \quad b(t) \le r \le 1. \tag{12}$$

Lemma 5 and $b' = f$ give the heat equations after the chain rule. Their interface values are zero, their slopes are p_-, p_+, and their slope difference equals $f = b'$. Choose the joining time $t_j = 3a/4$, strictly inside (t_*, a), and use the local temperatures until t_j and (12) afterwards. Lemma 6 identifies the two constructions on the open overlap (t_*, a). The interface defined in (11) also equals the local interface there. Thus the combined solution satisfies the PDE across the join with its initial data unchanged.

For $t \ge 2a$, all jets of f, p_-, p_+ vanish, so both series (12) are identically zero by Lemma 5, while $b = b_T$. Since $2a < T$, this gives the required terminal state. The exterior boundary controls are the traces $\theta_-(t,0)$ and $\theta_+(t,1)$; they vanish initially and are continuous later, hence bounded on $[0,T]$.

In the two fixed phase coordinates the combined temperatures are jointly continuous on compact time–space rectangles and vanish at the interface. Their zero extensions are therefore jointly continuous on $[0,T]\times[0,1]$: at a moving-interface crossing use continuity of the phase-coordinate profiles at their zero endpoint. The same argument at time zero uses the E_α continuity from Lemma 3 and the compact initial supports. Uniform continuity on this rectangle yields continuity into $L^2(0,1)$, since $\|u(t)-u(s)\|_2 \le \|u(t)-u(s)\|_\infty$. Return to the physical variables by

$$\ell(t) = Lb(t), \qquad y(t,x) = -\frac{\lambda L^2}{d_l}\theta_-(t,x/L), \qquad z(t,x) = -\frac{\lambda L^2}{d_r}\theta_+(t,x/L).$$

The boundary controls are $v_l(t) = -\lambda L^2\theta_-(t,0)/d_l$ and $v_r(t) = -\lambda L^2\theta_+(t,1)/d_r$. For every zero-extended normalized profile w, $\|w(\cdot/L)\|_{L^2(0,L)} = \sqrt{L}\,\|w\|_{L^2(0,1)}$. These fixed factors preserve boundedness and strong continuity, and the normalization calculation in (1) restores the physical Stefan law. This proves Theorem 1. □

AI-use statement. OpenAI Codex assisted in developing the argument, checking sources and preparing this manuscript. Independent mathematical review is requested.

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